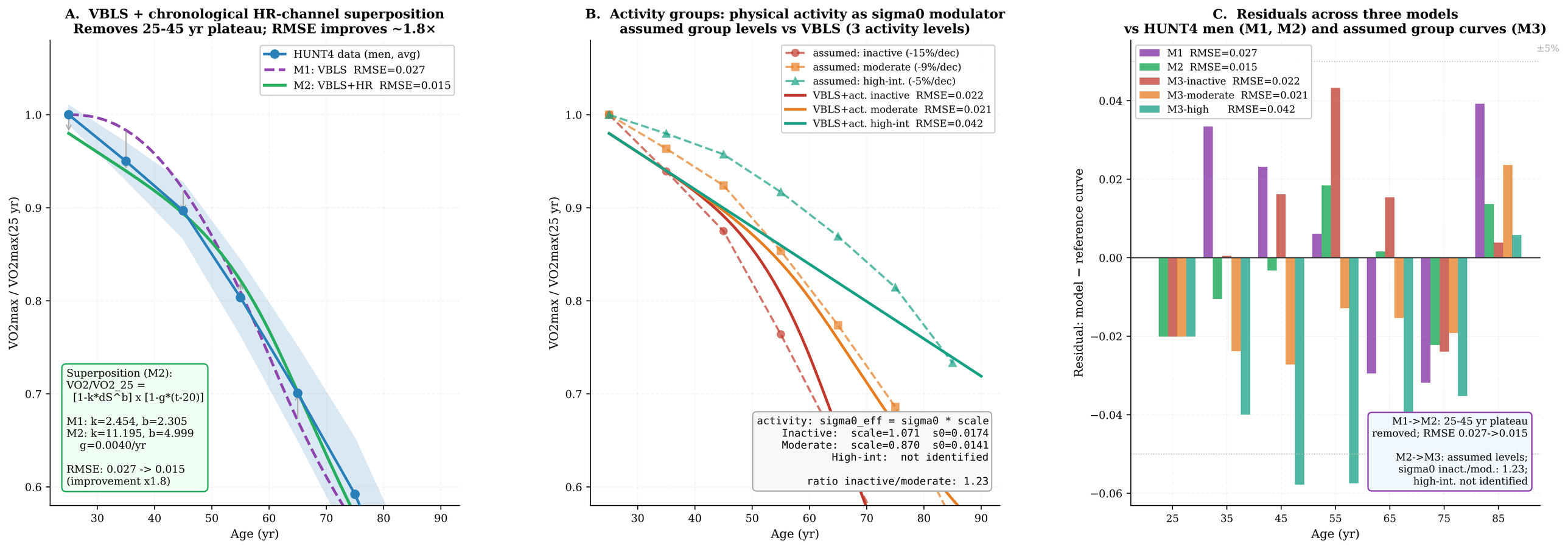

A. VBLS + chronological HR-channel superposition
Removes 25-45 yr plateau; RMSE improves ~1.8×
HUNT4 data (men, avg)
M1: VBLS RMSE=0.027
M2: VBLS+HR RMSE=0.015
Superposition (M2):
VO2/VO2_25 =
[1-k*dS^b] x [1-g*(t-20)]
M1: k=2.454, b=2.305
M2: k=11.195, b=4.999
g=0.0040/yr
RMSE: 0.027 -> 0.015
(improvement x1.8)
VO2max / VO2max(25 yr)
Age (yr)
B. Activity groups: physical activity as sigma0 modulator
assumed group levels vs VBLS (3 activity levels)
assumed: inactive (-15%/dec)
assumed: moderate (-9%/dec)
assumed: high-int. (-5%/dec)
VBLS+act. inactive RMSE=0.022
VBLS+act. moderate RMSE=0.021
VBLS+act. high-int RMSE=0.042
activity: sigma0_eff = sigma0 * scale
Inactive: scale=1.071 s0=0.0174
Moderate: scale=0.870 s0=0.0141
High-int: not identified
ratio inactive/moderate: 1.23
VO2max / VO2max(25 yr)
Age (yr)
C. Residuals across three models
vs HUNT4 men (M1, M2) and assumed group curves (M3)
M1 RMSE=0.027
M2 RMSE=0.015
M3-inactive RMSE=0.022
M3-moderate RMSE=0.021
M3-high RMSE=0.042
±5%
M1->M2: 25-45 yr plateau
removed; RMSE 0.027->0.015
M2->M3: assumed levels;
sigma0 inact./mod.: 1.23;
high-int. not identified
Residual: model − reference curve
Age (yr)